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PHYSICO-CHEMICAL ORIGIN OF SUPERPERMEABILITY – LARGE-SCALE EFFECTS OF SURFACE CHEMISTRY ON "HOT" HYDROGEN PERMEATION AND ABSORPTION IN METALS

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Superpermeability, the absorption of amounts of hydrogen much exceeding the equilibrium concentration, dependence of permeability upon characteristics of metal surface (for membranes of any thickness), and other particular effects, are exhibited when a metal membrane is acted upon by non-equilibrium "hot" hydrogen. The reason for this is that the process of thermal evolution of the dissolved, or implanted, hydrogen involves the associative step to form the diatomic molecule on the metal surface. This step may become effectively slow because of the presence of the activation barrier resulting, for transition metals, from non-metal adlayers on the surface. The effects of metal surface chemical composition on hydrogen uptake and permeation are demonstrated experimentally on the membranes of palladium, niobium and austenitic stainless steel interacting with hydrogen molecules, thermal atoms and fast ions. In the case of fast ions, hydrogen permeation and absorption increase, and re-emission decreases, by orders of magnitude, if chemically active gases (O_2, H_2O, H_2S or C_2H_2) are present in the vacuum system. The effect is accompanied by a deep minimum in the energy dependence of permeation and absorption corresponding to the maximum in the sputtering yield.

1. Introduction

We would like to raise the question of how the term superpermeability should be defined. In some authors' usage, the meaning of this word differs considerably from our original definition [1] suggested to denote a particular phenomenon that was exhibited in our experiments [2,3]. The term superpermeability is often referred to when *any increase* in metal permeability occurs as a result of the replacement of equilibrium hydrogen (i.e. hydrogen in thermal equilibrium with metal at a specified temperature) by "hot" hydrogen (atomized, ionized etc.). Such an effect does not seem so very singular and hardly requires so "emphatic" a term to be employed. According to our definition, superpermeability means that the permeation flux is comparable to the incident flux of gas particles or even nearly equals it. In our understanding, this term should point out approaching the total "transparency" of a membrane to incident gas flow, and should emphasize a sharp contrast to the common experience that solid membranes are impervious to gases (i.e. the permeation flux is orders of magnitude less than the incident one). Superpermeability, as we understand it, implies that the permeability of a membrane of usual thickness nears its conceivable limit: *the permeability of an opening of the same area*, and thus depends neither on metal temperature nor on the intrinsic bulk properties of the metal, "diffusion impedance" notwithstanding.

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2. Principal equations

2.1. Kinetics of dissolution and evolution

Hydrogen dissolution and evolution by metals should be considered as multi-step processes. Now, as it follows from a simple phenomenological treatment, superpermeability will be possible only when the rate of thermal evolution is controlled either by the associative step itself or by any post-associative step [4–6]. In this case, the process of thermal evolution is second order in subsurface concentration, c, of dissolved hydrogen, and the probability of thermal dissolution of a hydrogen gas molecule at one impact on a metal surface equals the probability of dissociative binding of the molecule, α. Correspondingly, the rates of thermal dissolution, j_a,

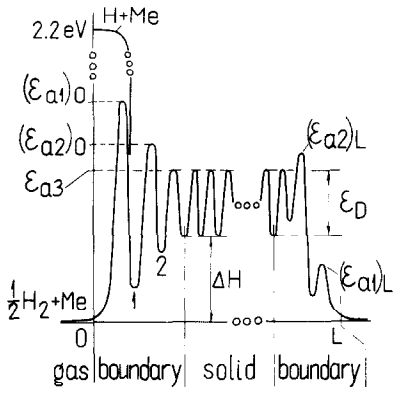


Fig. 1. Potential energy diagram for the hydrogen–metal system. The energy of the H atom in the H_2 molecule in the gas phase is taken as the zero energy level. The step $1 \rightarrow 0$ suggests the association of two H atoms. The energy level $\mathcal{E}_{a2}$ indicates the barrier of the slowest step among pre-associative ones in the multi-step process of thermal hydrogen release from the metal, and $\mathcal{E}_{a1}$ indicates the slowest in the series of steps which include the associative step and all post-associative steps. This potential relief is attributed to the portion σ of the metal surface occupied by those active centres, essentially responsible for thermal dissolution/evolution. The other sites on the surface will display larger barriers $\mathcal{E}_{a1}$.

and evolution, j_r, are given (at a condition that the surface coverage by hydrogen is far from saturation) by

$$j_{a0} = \alpha_0 Z p, \qquad j_{r0} = k_{r0} c_0^2, \qquad j_{aL} = \alpha_L Z p, \qquad j_{rL} = k_{rL} c_L^2, \tag{1}$$

where p is hydrogen pressure, Z is the kinetic theory coefficient ($Z = (2\pi mkT)^{-1/2}$, where m is the mass of the gas molecule, and k is Boltzmann's constant; $Z \approx 1.2 \times 10^{19}$ $H_2/(cm^2\ s\ Pa)$ at 20 °C), k_{r0} and k_{rL} are the rate constants for evolution, with indices 0 and L referring to the inlet (upstream) and outlet (downstream) surfaces, respectively. The probabilities α may be presented as

$$\alpha_0 = \alpha_0^* \sigma_0 \exp[-2(\mathcal{E}_{a1})_0/RT], \qquad \alpha_L = \alpha_L^* \sigma_L \exp[-2(\mathcal{E}_{a1})_L/RT]; \tag{2}$$

the barriers $\mathcal{E}_{a1}$ are shown in fig. 1. The quantities α_0^* and α_L^* are pre-exponentials in the order of 0.1–1 while σ_0 and σ_L are fractions of the surfaces occupied by the centres active in the dissolution-evolution kinetics considered # (for a homogeneous surface $\sigma = 1$).

The remaining part $1 - \sigma$ of the surface has a larger $\mathcal{E}_{a1}$ barrier.

From balancing the fluxes, it follows that $k_r = \alpha Z/S^2$, where S is the equilibrium (Sieverts') constant for dilute solution of hydrogen in the metal contacting gas phase hydrogen:

$$S = S^* \exp(-\Delta H/RT); \tag{3}$$

ΔH is the enthalpy of solution (see fig. 1, and the table of notation in the Appendix).

2.2. Permeation and steady-state concentration

We shall neglect in the further treatment non-thermal mechanisms of re-emission (here we refer to the ion-induced desorption in the case of fast ions, and to recombination according to the Eley-Rideal mechanism in the case of thermal hydrogen atoms) *. The flux entering the metal lattice will then be expressed via the incident "hot" gas flux j_i as ξj_i. The permeation probability χ for a "hot" gas particle, per single impact upon the membrane surface, and the steady-state permeation flux j_{rL}, may now be presented [4,16] as

$$\chi = \min \begin{cases} \xi \dfrac{\alpha_L}{\alpha_0 + \alpha_L}, & \text{(4a)} \\ \dfrac{SD}{L}\sqrt{\dfrac{\xi}{j_i Z \alpha_0}} = \dfrac{S^* D^*}{L}\sqrt{\dfrac{\xi}{j_i Z \alpha_0^* \sigma_0}} \times \exp\dfrac{(\mathcal{E}_{a1})_0 - \mathcal{E}_{a3}}{RT}; & \text{(4b)} \end{cases}$$

$j_{rL} = \chi j_i$.

Here $D = D^* \exp(-\mathcal{E}_D/RT)$ is the diffusion coeffi-

* For clean metal surfaces, the non-thermal re-emission may obviously be neglected when the surface coverage by hydrogen θ is small [5,7,8]. Given feasible amounts of the incident flux, we shall have for most metals $\theta \ll 1$ on clean surfaces at temperatures above 500 K [9]. Evaluating the importance of non-thermal re-emission machinisms is more complicated for the surfaces covered by non-metal contaminant layers. The concentration of chemisorbed hydrogen on the surface is negligible in this case, but the subsurface concentration of dissolved hydrogen may be increased significantly because of the impeded thermal evolution. Anyway, Eley-Rideal desorption was never observed in our experiments [10–15] with non-metal-covered Pd, Ni and Fe acted upon by thermal H atoms at temperatures as low as 300 K. Replacing the atoms by fast ions often results in a substantial re-emission rise, but our experience is that, at least at higher metal temperatures, this rise is usually owed to the thermal evolution getting faster due to the surface cleaning. So we tend to think that non-thermal re-emission may again be ignored in most cases at temperatures above 500 K for the surfaces covered by non-metallic elements.

Table 1
Several hydrogen–metal system characteristics

	Fe [17]	Ni [17]	Pd [17]	Nb [17]	V [17]	Austenitic stainless steel (ASS) [18]
D^* (10^{-3} cm^2/s)	0.93	6.7	5.2	0.5	0.35	85
S^* (10^{17} (atom H) cm^{-3} Pa$^{-1/2}$)	2.4	2.6	3.3	0.61	1.0	2.4
$\mathscr{E}_{a3}$ (kJ/mol H)	36	54	15	−28	−28	67
ΔH (kJ/mol H)	24	17	−10	−39	−33	8.8

cient; the barriers $\mathscr{E}_{a1}$, $\mathscr{E}_{a3}$ and $\mathscr{E}_D$ are shown in fig. 1. Note that eq. (4a) may not be precisely accurate if $\alpha_L \gg \alpha_0$ *. For a symmetric membrane, with $\alpha_0 = \alpha_L$, eq. (4a) yields

$$\chi = \tfrac{1}{2}\xi. \tag{4c}$$

The characteristic transient time τ to attain the steady state permeation is given by [1,10,15]

$$\tau = \max\begin{cases} \dfrac{LS}{\sqrt{2Z\xi j_i(\alpha_0+\alpha_L)}} = \dfrac{L}{\sqrt{2\xi j_i(k_{r0}+k_{rL})}}, & (5a) \\ \dfrac{L^2}{6D} = \dfrac{L^2}{6D^*}\exp\dfrac{\mathscr{E}_D}{RT}. & (5b) \end{cases}$$

The quantities S^*, D^*, $\mathscr{E}_{a3}$ and ΔH for several metals are summarized in table 1. The quantities α_0 and α_L are determined by σ and $\mathscr{E}_{a1}$ values according to eq. (2). σ values ranging from 10^{-3} (refs. [6] and [10]) to 1 (refs. [7,9,19–22]) are known from experiments as well as $\mathscr{E}_{a1}$ ranging from 0 ** to ~40 kJ/mol H (refs. [1,7,10] and [23]). The physico-chemical nature of these extremely wide-ranging variations of σ, $\mathscr{E}_{a1}$ and, consequently, of α, is addressed by section 3 of the present paper.

Superpermeability, in our terms, implies

$$\chi = \text{const}(T) = 0.1\text{–}1. \tag{6}$$

* Eq. (4a) is reduced to $\chi \approx \xi(1-\alpha_0/\alpha_L)$ in such a case, but this holds only as long as $\alpha_L \ll S^2D^2/(Z\xi L^2 j_i)$ – cf. eq. (9). If the reverse inequality is valid, the χ value will be approaching ξ according to another low [7]: $\chi \approx \xi(1-k_{r0}\xi L^2D^2j_i)$. This difference should be taken into account when the magnitude of the probability of re-emission, $\chi - \xi$, is important.

** The dissociative chemisorption of molecular hydrogen on transition metals requires no activation energy [9]. As for the net process of hydrogen dissociative dissolution through the transition metal clean surfaces, it has been shown to go with no activation energy in several direct experiments [14,20,21].

For this to take place, three simultaneous conditions are necessary, as follows from eqs. (4),

$$\xi = \text{const}(T) = 0.1\text{–}1, \tag{7}$$

$$\alpha_L \geqslant \alpha_0, \tag{8}$$

$$L\sqrt{j_i} \ll \frac{SD}{\sqrt{Z\xi}}\frac{1}{\sqrt{\alpha_0}} = \frac{S^*D^*}{\sqrt{Z\xi}}\frac{1}{\sqrt{\alpha_0^*\sigma_0}}\exp\frac{(\mathscr{E}_{a1})_0 - \mathscr{E}_{a3}}{RT}. \tag{9}$$

Eq. (7) is the condition of free passage into the metal lattice and it is satisfied, if the gas is "hot" enough. The meaning of condition (8) is clear, and we shall address it further on when discussing experimental results. Eq. (9) is the condition of "strong reflection" [4,16] of dissolved hydrogen atoms from the inlet boundary. It means that the inlet (upstream) boundary of a membrane impedes the dissolved atoms recombinative desorption to such an extent that essentially every atom reaches the downstream boundary by way of interstitial random-walk diffusion rather than being released through the inlet surface when first approaching it. A high barrier $(\mathscr{E}_{a1})_0$, a small portion σ of the surface accessible to the thermal release of the gas, and a sufficiently small incident flux j_i may contribute to satisfying this condition. Dependence on j_i appears here because of the associative character of the evolution process. If a condition similar to eq. (9) is also valid for the downstream surface, then the dissolved atoms will cross the membrane repeatedly; eq. (5a) gives the mean time of stay inside the membrane for a dissolved atom.

For the molecular hydrogen at thermal equilibrium, assuming $\xi = \alpha_0$ and $j_i = Zp$ in eqs. (4), we get

$$\chi = \min\begin{cases} \dfrac{\alpha_0\alpha_L}{\alpha_0+\alpha_L}, & (10a) \\ \dfrac{SD}{L}\dfrac{1}{Z\sqrt{p}} = \dfrac{S^*D^*}{L}\dfrac{1}{Z\sqrt{p}}\exp\left(-\dfrac{\mathscr{E}_{a3}}{RT}\right); & (10b) \end{cases}$$

for a symmetric membrane [4,24], with $\alpha_0 = \alpha_L = \alpha$, eq. (10a) reduces to

$$\chi = \tfrac{1}{2}\alpha \tag{10c}$$

Permeation in the regime $\chi \propto 1/L$ is often spoken of as "diffusion controlled", which seems to us somewhat unlucky. Such a term ought to point up that surface processes have no effect upon permeability, but this applies to an equilibrium gas only (eq. (10b)), whereas for a "hot" gas, permeability indispensably depends on the characteristics of upstream surface (eq. (4a)): the permeation probability, χ, increases with $(\mathscr{E}_{al})_0$ rising and σ_0 diminishing (e.g. due to passivation), in the $\chi \propto 1/L$ regime.

The steady state concentration in a sample of metal plunged into a "hot" gas essentially depends upon the surface properties too, in contradistinction to equilibrium concentration $c_e = S\sqrt{p} = S^*\sqrt{p}\exp(-\Delta H/RT)$. Balancing the fluxes ($j_i = k_r c^2 = (Z\alpha/S^2)c^2$) and using eqs. (2) and (3), one obtains

$$c = \sqrt{\frac{j_i \xi}{Z\alpha}}\, S = \sqrt{\frac{j_i \xi}{Z\alpha^*}}\, S^* \frac{1}{\sqrt{\sigma}} \exp\frac{\mathscr{E}_{al} - \Delta H}{RT}. \tag{11}$$

This topic is discussed elsewhere [4,10,25], and is dealt with in detail in subsequent contributions of us [5,7] and by other workers [6,13,23,26–28].

2.3. Further comments

The implantation depth, l, for "hot" gas particles is not included in eqs. (4)–(9) and (11), which is relevant to the situation when the back "reflection" of dissolved hydrogen atoms effectively ensures their passing inside the lattice to a greater depth than l.

Expressions equivalent or close to eqs. (4), (5) and (11) have been also given by Waelbroeck et al. [6,29], by Timashov et al. [30,31] and by Pick et al. [27,28]. The fit of these expressions to experimental data can be verified by comparing the results of several independent experiments. Suppose it is shown in an experiment in the superpermeability regime [1,6,10,13,15,32] on a symmetric membrane ($\alpha_0 = \alpha_L = \alpha$) that the transient time, τ, depends on metal temperature otherwise than $L^2/6D$ does, that $\tau \gg L^2/6D$, and $\tau \propto j_i^{-1/2} L$. This is likely to suggest the validity of eq. (5a), and also of eqs. (4c) and (10c). Now one can try to derive the quantity α from the data on permeability to molecular hydrogen by employing eq. (10c). Substituting this α in eqs. (5) and (11), we shall predict the expected values of τ and c, as well as their temperature dependencies. A good agreement is usually observed between such computations and the experimental data for thermal hydrogen atoms [1,6,10,11,13,15,32]. The same pertains to the evaluation by eq. (9) of the ranges of L, j_i and metal temperatures where superpermeability to atomic hydrogen should take place.

All possible expressions for χ and c are not covered by eqs. (4) and (10), since these are valid when the law $j_r = k_r c^2$ for gas evolution is obeyed. However, the evolution may follow the law $j_r \propto c$ over real ranges of j_i, T, α_0 and α_L for many metals and alloys (particularly, for austenitic stainless steel), and another pattern of dependence of χ and c on T, L, j_i, ion energy and surface characteristics will emerge [7,33]. Still in the present paper we treat only the case $j_r = k_r c^2$, as it enables reaching or approaching superpermeability.

2.4. Non-symmetric membranes

Seen from eqs. (4) is that a non-symmetric membrane ($\alpha_0 \neq \alpha_L$) may exhibit conspicuous differences in the permeability to "hot" gas for the two opposite directions of gas flow (i.e. for α_0 and α_L interchanged). This trait is peculiar to non-equilibrium conditions, whereas the permeability to the molecular hydrogen gas is the same in either direction for both the extremes given by eqs. (10a) and (10b). We have to dwell on the latter point, since arguing about it resumes intermittently in the literature [34,35].

Permeation differences in the different directions are not entirely prohibited for a non-symmetric membrane separating an equilibrium gas at some specific pressure from the vacuum, or from the gas at another pressure, if more complicated mechanisms of hydrogen–metal interaction are involved than those assumed to obtain eqs. (4) and (10) *. But any kinetic formalism being contemplated to account for an asymmetric permeation should meet one requirement stipulated by the principles of thermodynamics: no regular gas flow may exist under isothermic conditions, if hydrogen pressures are equal on the both sides of a membrane.

This requirement appears to be disregarded by alleging that the deposition of palladium overlayers only on the upstream surface of the originally symmetric membranes of stainless steel [37] or niobium [38] should bring about a sharp rise of their permeability to molecular hydrogen owed simply to the easier hydrogen dissociation and dissolution characteristic of palladium. A similar argument [34] will infer a non-zero gas flow, if

* To be accurate, the difference is possible even in this case up to about 60% for the permeation regime transitional from eq. (10a) to eq. (10b) [6,36].

applied to the situation of equal pressures at either side of the asymmetric membrane. Again, the possibility of easier dissolution on the upstream side is included in eq. (10a). The quantities α_0 and α_L, pertaining to the two membrane sides, are equipollent there, and a two-fold permeation increase maximally may result from enhancing the rate of dissolution at only one side of a symmetric membrane.

A notice should be given to one possible source of mistake in assessing the data of experiments with molecular hydrogen. In a real experimental set-up, the temperature of the gas acting on a metal sample may differ from the metal temperature. The sample can be heated, while the gas may retain essentially the temperature of the walls of vacuum chamber. The effects may develop in these circumstances, characteristics of non-equilibrium conditions. For example, the probability of absorption of a hydrogen molecule at the surface of copper is known to depend on the molecule kinetic energy [39,40], whereas clean palladium absorbs the impinging molecules with a probability of the order of unity, irrespective of their velocity [20,40]. A palladium membrane having a very thin copper coating on one surface may, therefore, display an asymmetric permeability to a gas that is "colder" than the membrane. The permeation will be smaller when the gas acts on the copper-covered surface which repulses the slow molecules falling from the gas. The asymmetry is to disappear on equalizing the gas and the metal temperature.

3. Small values of the probability of thermal dissolution α: interpretation versions

The capacity to "lock in" the dissolved hydrogen, which gives rise to superpermeability and related phenomena, is governed for any particular metal surface by the quantity α, i.e. by the level of barrier $\mathscr{E}_{a1}$ (fig. 1) and by the portion of surface σ accesible to thermal dissolution/evolution, as it follows from the above treatment (eqs. (2), (4), (5), (9) and (11)). A variety of interpretations exist as regards the physics behind the large $\mathscr{E}_{a1}$ and small σ values displayed in many experiments. These viewpoints may be classified into two categories.

3.1. Version I: attributing to clean surface properties

The first interpretation, going back to the classical works by Günterschulze, Betz and co-workers [41,42], ascribes the small values of α to the fundamental properties of hydrogen–metal systems. The high barrier to hydrogen dissolution and the small number of active centres is regarded characteristic of clean metal surfaces; e.g. for iron, the figures $\mathscr{E}_{a1} = 22$ kJ/mol H and $\sigma \approx 3 \times 10^{-3}$ have been specified [6,29], which easily allows satisfying eq. (9).

Applying this concept consistently faces a certain difficulty, as the values of $\mathscr{E}_{a1}$ and σ like those mentioned should be brought into line with the kinetics of dissociative chemisorption which is known to occur on transition metal surfaces with the sticking probability of 0.1–1 and with no activation energy [9]. If (i) the transfer of hydrogen from the gas phase into solution goes through the states of dissociative chemisorption, and (ii) eqs. (1) are valid pointing out that either some pre-dissociative step or the dissociative step is rate-limiting in the dissolution kinetics, then virtually any atom chemisorbed goes into solution in the bulk of the metal. Consequently $\alpha = 0.1\text{–}1 = \text{const}(T)$, $\mathscr{E}_{a1} \ll RT$ and $\sigma \approx 1$ should have been observed, and not $\alpha \ll 1$, $\mathscr{E}_{a1} \gg RT$ and $\sigma \ll 1$.

One could avoid this discordance by assuming the solubilization to occur bypassing the usual chemisorption states [43,44], so that the values of $\mathscr{E}_{a1}$ and σ revealed might be ascribed to a bypassing process. But then it is to be explained, why the seemingly slowest of the two alternative mechanisms (with $\alpha \ll 1$) comes into effect. One may think the transition out of the deep chemisorption well into the metal lattice to be actually "frozen" at sufficiently low temperatures, so that the flux from the surface into the bulk metal becomes much smaller than the flux falling upon the surface from the gas. Still superpermeability has been observed over wide ranges of temperature, including high ones, up to 500–1500 K [1,6,11–14,19,23,32,45–48]. Simple estimates [5,6,36] show that to prevent effectively chemisorbed atoms from entering the bulk of the metal, the level, $\mathscr{E}_{a2}$, of the barrier which separates the chemisorption sites from interstitial sites inside the lattice should be much higher than the level $\mathscr{E}_{a3}$ (see fig. 1).

That a possibility exists for the process of hydrogen evolution (hence, for dissolution as well) to bypass chemisorption states is to be assumed from the very interesting results by Comsa et al. [39,49] on angular and energetic distributions of hydrogen molecules released from clean surfaces of Cu and Ni(111). Nevertheless, the total probability of dissociative dissolution α (integrated over the whole solid angle), if derived from these data, would not be small enough to satisfy eq. (9) and thus to give rise to superpermeability.

3.2. Version II: attributing to surface impurities

The experiments on clean surfaces are required to prove whether or not the intrinsic fundamental properties of a transition metal may actually result in $\mathscr{E}_{a1} \gg RT$ and $\alpha \ll 1$, and give rise to superpermeability. But then, in many experiments both the treatment applied to clean the surface and the procedures to monitor surface cleanliness seem utterly inadequate as far as researches dealing with hydrogen uptake, permeation and release are concerned. The example of palladium is very typical. It is established [20,50] that H_2 dissolution in Pd occurs, contrary to the previous data [24,43,44], via the usual chemisorption states, so that $\alpha = 0.3\text{–}1 = \text{const(T)}$, and hence $\mathscr{E}_{a1} \ll RT$ and $\sigma \approx 1$. The same is found for niobium [21,22] as long as the surface is scarcely covered by adsorbed hydrogen.

An alternative standpoint to that discussed above had been used to explain why the probability of dissolution, α, might be so small. Put forward in all our contributions, and by some other workers [26,27,51,52], is the idea that it is the passivation of surface by non-metal contaminants ("surface valence bonds saturation" according to Tontegode and co-workers [53,54]) that entails, at least for transition metals, α becoming very small and strongly dependent on temperature ($\alpha \ll 1$, $\mathscr{E}_{a1} \gg RT$ and $\sigma \ll 1$). This results in superpermeability with respect to a "hot" gas.

Even monolayers of non-metallic elements on metal surfaces are known to cause orders-of-magnitude decrease in the rates of catalytic processes involving dissociation of diatomic molecules [55,56]. Also, the deposition of non-metallic monolayers causes drastic changes in the energy and angular distributions of hydrogen molecules released from transition metals. The fast molecules appear in the energetic spectra, desorbed in narrow solid angles close to the normal [40].

We are going to demonstrate the large-scale effects of surface chemistry on the uptake, permeation and re-emission, when a metal is acted upon by "hot" hydrogen particles, and the ways to control these effects.

4. Experiments with thermal atoms

4.1. Incompletely cleaned membrane surfaces

Our earliest experiments were purposely performed on membranes whose surfaces had been incompletely cleaned, as we have been contending that the small values of the hydrogen molecules' thermal dissolution probability, α, was due to surface passivation by residual non-metal contaminants [4,10,11]. We also employed the "softest" kind of "hot" gas, i.e. hydrogen thermal atoms, lest the surface contaminant layers be destroyed. These layers remain undamaged under the action of thermal atoms over the whole range investigated of metal temperatures and incident atomic fluxes (up to 10^{15} cm^{-2} s^{-1}), which was proven for all the investigated metals, Pd (refs. [10,11,13] and [32]), Ni (refs. [1] and [13]), Fe (refs. [12] and [13]), Nb (refs. [45] and [47]) and austenitic stainless steel (ref. [46]), by means of physico-chemical tests, such as H_2 and CO adsorption [14,19], kinetics of H_2 dissolution [1,10–14, 19,32,45,47], kinetics of catalytic decomposition of C_2H_2, C_6H_6 and H_2S, mass-spectrometric monitoring of the formation of CO, CO_2 and SO_2 during the gasification of contaminants (C and S) by heating in oxygen [13,14,19].

When the H_2 thermal dissolution probability, α, was measured as a function of temperature, $\alpha \ll 0.1$ and $\mathscr{E}_{a1} \gg RT$ was always observed, and $\sigma \ll 1$ more often than not, with no palpable changes in these quantities detected in the course of experiments with atomic hydrogen. A membrane, if originally symmetric ($\alpha_0 = \alpha_L$), remains so, even though only the upstream side is subject to the action of atomic hydrogen. But when the surface is effectively cleaned *, there are dramatic changes in the kinetics of H_2 dissolution (becomes $\alpha = 0.1\text{–}0.6 = \text{const}(T)$, $\mathscr{E}_{a1} = 0$ and $\sigma = 1$), and in the other "testing" processes mentioned above, at least for those metals (Pd – refs. [20] and [50], and Nb – refs. [21] and [22]) for which such measurements have been carried out.

It is the incompletely cleaned surfaces that lead to superpermeability of palladium [10,11,13,32], nickel [1,13], iron [12,13] and niobium [45,47] interacting with atomic hydrogen. One essential point should be noted here. The very fact of superpermeability ($\chi = 0.1\text{–}1 = \text{const}(T)$) suggests that the probability of dissolution of thermal atoms per single impact upon the surface, ξ, is large: $\xi = 0.1\text{–}1 = \text{const}(T)$. Were the relations $\mathscr{E}_{a1} \gg RT$ and $\sigma \ll 1$ intrinsically characteristic of the metal surface, as associated with breaking the chemical bond in H_2 molecule, such a free transfer of hydrogen, when atomized in advance in the gas phase, into the metal

* Palladium can be baked at moderate temperatures in the presence of oxygen at low pressures to "burn out" the impurities. Niobium may be heated to 2500 K in ultrahigh vacuum, but the bombardment by H^+ and Ar^+ ion beams was utilized in our experiments.

lattice would seem only natural (cf. fig. 1). This is not so evident even in the case of monatomic non-metal layer, as the latter inhibits the dissolution of H_2 molecules quite effectively ($\alpha \ll 0.1$). The unhampered passage of thermal hydrogen atoms through the non-metal layers and into the bulk of the metal manifests itself in all our experiments, and we believe that the data by Waelbroeck and co-workers [6,23,29,48] should be interpreted in the same way. Apparently, this mechanism is typical to the metal surfaces coated by non-metallic films of the thickness close to one monolayer.

4.2. Controlled chemical composition of membrane surfaces

As the peculiarity of the characteristics of the passivated metal surfaces became clear, it was logical to turn to the experiments in which chemical composition of the surface was definite. The monolayers of carbon and sulphur were deposited onto the clean ($\alpha = 0.3$–$0.6 = \text{const}(T)$) upstream and downstream surfaces of a palladium membrane by means of controlled pyrolysis of acetylene [19] and hydrogen sulphide [14] in situ, inside the UHV experimental apparatus. The C and S monolayers are quite stable both in molecular and atomic hydrogen within the whole range of metal temperatures and incident fluxes investigated. Firstly, this provides the opportunity to observe superpermeability in experiments with atomic hydrogen, and, secondly, demonstrates that the usual treatment of metal surfaces in hydrogen is absolutely insufficient to clean them.

Shown in fig. 2 are the data of permeation experiments performed on 20 μm-thick Pd membranes with monolayers of carbon [19] or sulphur [14] on either surface. One can see the following:

(1) For the interaction with molecular hydrogen, the permeation probabilities χ, while differing substantially from one another in the cases of C- and S-covered Pd, are orders of magnitude less than for clean palladium [20]. These permeability data yield the probability of H_2 dissolution, α, since, in agreement with eq. (10c), $\alpha = 2\chi$. One gets $\alpha = 0.15 \exp(-16.8 \text{ kJ mol}^{-1}/RT)$ ($\sigma \approx 1$, $\mathscr{E}_{a1} = 8.4$ kJ/mol H) for C-covered Pd, and $\alpha = 0.1 \exp(-34.4 \text{ kJ mol}^{-1}/RT)$ ($\sigma \approx 1$, $\mathscr{E}_{a1} = 17.2$ kJ/mol H) for S-covered Pd, while for clean Pd $\alpha = 0.3$–$1 = \text{const}(T)$ is known [20,50]. The barriers $\mathscr{E}_{a1}$ do "reflect back" dissolved atoms effectively enough, so that eq. (9) for a Pd membrane of thickness, L, of 20 μm at a temperature, T, above 300 K will be satisfied for the following incident fluxes: $j_i \lesssim 10^{15}$ cm^{-2} s^{-1} for C–Pd–C, and $j_i \lesssim 10^{17}$ cm^{-2} s^{-1} for S–Pd–S. If hydrogen thermal atoms are able to penetrate freely the C or S overlayers on Pd, superpermeability should take place at the values of L, T and j_i as indicated above.

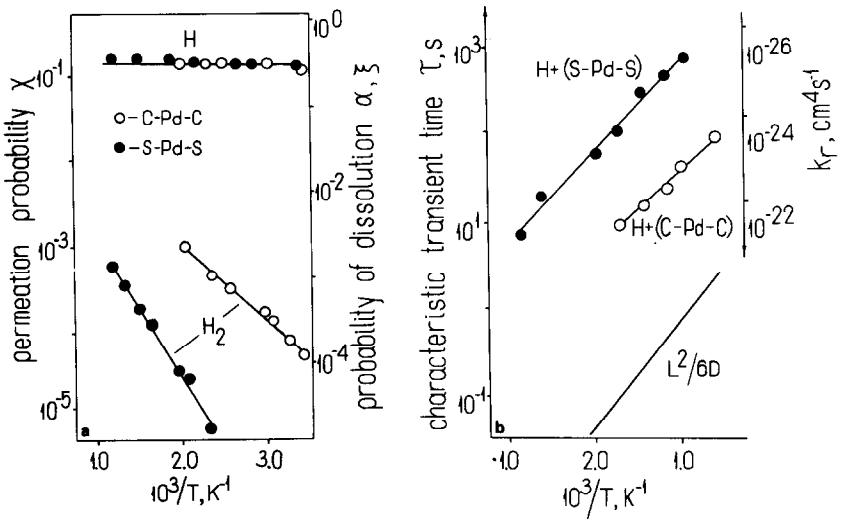


Fig. 2. Hydrogen permeation through a 20 μm-thick Pd membrane passivated symmetrically by C and S monolayers: the data on interaction with thermal H_2 molecules and H atoms [14,19]. The permeation probability χ here does not depend on the incident flux j_i for H_2 molecules as well as for H atoms. The characteristic transient time τ is proportional to $j_i^{-1/2}$, the data presented being obtained for $j_i = 6 \times 10^{13}$ H atoms/cm^2 s. The temperature of the C–Pd–C membrane was never raised above 500 K, because the monolayer of C dissolves rapidly in Pd metal at higher temperatures [19]. The dots on the plots denote the experimental values for χ and τ. The scales for α, ξ and k_r are derived from χ and τ according to eqs. (4c), (5a) and (10c).

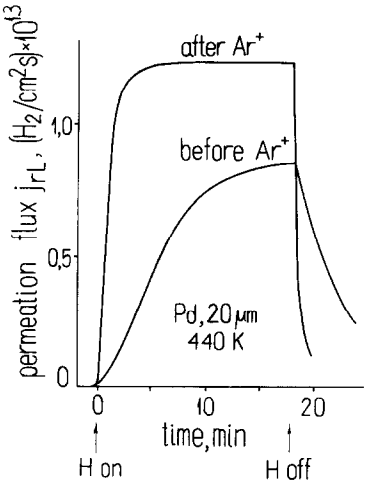


Fig. 3. Effect of cleaning downstream surface on the steady-state permeation flow and on the time of its rise and decay. The two curves show two succesive runs on the same 20 μm-thick Pd membrane: (i) when both the surfaces are passivated symmetrically by residual contaminants due to incomplete cleaning, and (ii) following cleaning of the downstream surface by Ar^+ ion bombardment (900 eV, 20 $\mu A/cm^2$, 1 min). Note that the steady state permeation did not exactly double (the increment here is slightly less than two-fold), because an undesigned initial passivation hardly ever is exactly symmetric.

(2) For the interaction with atomic hydrogen, superpermeability ($\chi \approx 0.1 = \text{const}(T)$) has been observed in the whole range of temperatures investigated for C–Pd–C as well as for S–Pd–S membranes. Thus the dissolution of thermal atoms through the monolayers of C and S occurs with probability of $\xi = 2\chi = 0.2 = \text{const}(T)$ (cf. eq. (4a)).

(3) The transient time τ is larger and its temperature dependence is stronger for S–Pd–S than for C–Pd–C, which corresponds to the higher barrier $\mathscr{E}_{a1}$ in the former case (cf. eqs. (2) and (5a)). In both cases $\tau \gg L^2/6D$. The experimental data follow the expected dependence $j_i^{-1/2}$. The correlation between the numerical values of τ and α obtained experimentally is also in reasonable agreement with eq. (5a), allowing for the divergence of Sieverts' constant values.

That superpermeability is governed by surface chemistry and by the kinetics of surface processes may be demonstrated rather convincingly by the transition from a symmetric to a non-symmetric passivation when the downstream membrane surface is cleaned in situ in the course of the experiment, as shown in fig. 3. With the barrier $(\mathscr{E}_{a1})_L$ vanishing during the cleaning treatment, the time, τ, for hydrogen atoms to stay inside the membrane falls by orders of magnitude, and the steady state permeation increases by approximately a factor of two. This complies with eqs. (4a) and (5) when $\alpha_L = \alpha_0$ changes to $\alpha_L \gg \alpha_0$.

The methods of passivation as set forth above may afford an important opportunity to manufacture the superpermeable membranes. One sees from fig. 2 that passivation by sulphur seems more advantageous than by carbon in that the usual permeability, with respect to molecular hydrogen, is much more suppressed. Hence, the larger degree of compression is attainable, if using such membranes to pump out and recuperate hydrogen [4,6,16,32,48,57]. Also in this case, superpermeability may keep up at larger values of $L\sqrt{j_i}$, as seen from eq. (9).

4.3. Hydrogen concentration in metal

Steady state concentration of dissolved hydrogen in a symmetric membrane ($\alpha_0 = \alpha_L$) is virtually coordinate independent when superpermeability takes place. The concentration, c, is directly linked [4,15] with the transient time τ: from eqs. (4a), (5a) and (11) it follows that $c = \tau(4\chi j_i/L)$, which also agrees with experimental data. The meaning of this relation is quite clear

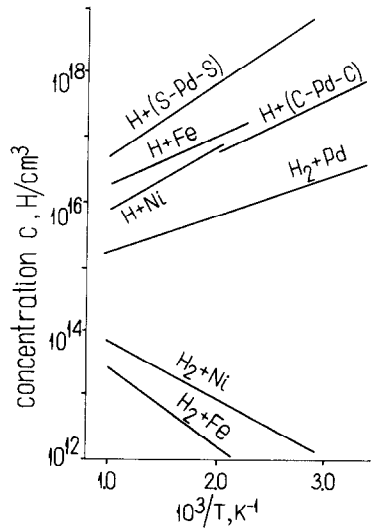


Fig. 4. Steady state hydrogen concentration in several metals acted upon by thermal molecules (equilibrium [17]) and by thermal atoms. The metal surfaces are passivated: for Fe and Ni specimens, by residual contaminants [13], and for Pd, by carbon and sulphur monolayers deposited symmetrically on either surface [58]. The incident flux j_i is 6×10^{13} cm^{-2} s^{-1}.

[4,6,10,15], because τ is the time of filling up the membrane with hydrogen. The concentration, c, can be derived from the values τ and χ measured, which is done in fig. 4.

The gas content in a metal acted upon by "hot" gas depends upon the state of the surface to a much greater extent than upon individual "bulk" properties of the metal. As a result, the relative ability of different metals to absorb hydrogen may not only deviate from the usual, but even become the reverse. Thus, passivated nickel, as seen in fig. 4, absorbes more hydrogen than the carbon-covered palladium does, and, in turn, the passivated iron exceeds the passivated nickel when these specimens are interacting with atomic hydrogen. A stronger passivation, by sulphur, puts palladium back at the top in ability to take up atomized hydrogen. It should be recognized that for a clean surface, the difference between atomic and molecular hydrogen may vanish [4,26] (with $\alpha = \xi \approx 1$ in eq. (11), one will get $c = c_e$).

5. Experiments with fast ions

5.1. Austenitic stainless steel

One might think that, if superpermeability manifests itself in experiments with thermal atoms, then it is even more likely to take place for fast ions implanted within the bulk of the metal. Yet rather the opposite follows from the physico-chemical model outlined here. First, the depth of implantation may be of no importance at all, as in eqs. (4)–(9). Secondly, the surface contaminant films responsible for $\mathscr{E}_{al} \gg RT$ and $\sigma \ll 1$ may be purged by ion sputtering. The elimination of the contaminants from the upstream surface would result in a sharp decrease in the permeation and in the steady state concentration when the thermal atoms are replaced by the fast ions.

This is just the case for austenitic stainless steel (ASS) [46,51,52,59,60] (see, for example, fig. 5), and also for iron [61,62], nickel [63], molybdenum [63] and niobium (for the latter see below).

Such effects might be tentatively interpreted as due either to changes in the diffusion coefficient caused by the ion bombardment [63], or, in line with Betz's ideas [42], to the formation of some specific centres of thermal desorption (recombination) on the genuinely clean metal surface by ion bombardment [6,62]. Still it is the contaminants that appear to govern the effect, as evidenced by our experiments in which active gases were admitted onto the upstream side of a membrane acted upon by hydrogen ion beam. Figs. 5–10 are presented to illustrate this.

The admittance of oxygen at a pressure as low as 4×10^{-4} Pa onto a ASS membrane at 940 K enhances

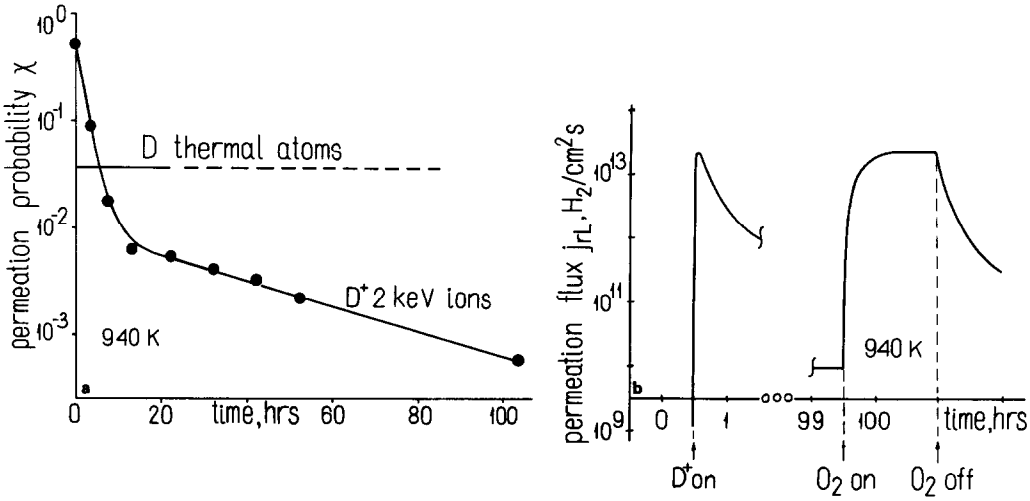


Fig. 5. The drift of the steady-state ion driven permeation through a 100 μm-thick austenitic stainless steel membrane during the prolonged action of the D^+ ion beam (2 keV, $j_i = 3 \times 10^{14}$ D^+/cm^2 s) in comparison with the stationary permeation under the action of the thermal D atomic beam (a). Transient permeation after initiating the D^+ beam and after the admission of oxygen at 4×10^{-4} Pa onto upstream side of the membrane (b). The preliminary cleaning procedure consisted of heating the membrane to 1020 K in an ultrahigh vacuum. Data from ref. [59].

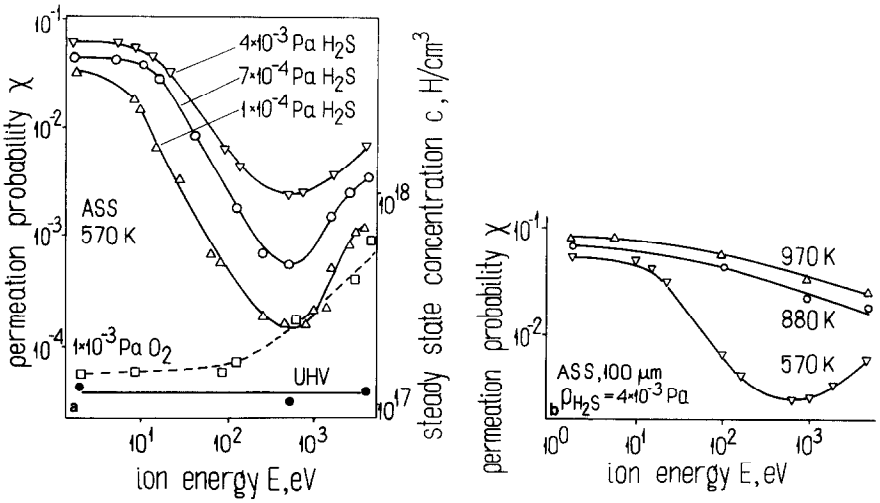


Fig. 6. Effect of chemically active gases present on upstream side of an austenitic stainless steel membrane upon permeation probability χ (experimental) and upon steady state concentration of hydrogen c (deduced from χ). The plots represent the dependence of χ and c on acting ions energy E. Varying from one curve to another are: the composition of gas atmosphere and the pressure of the gases (a); membrane temperature (b). The incident flux j_i is 3×10^{14} cm^{-2} s^{-1}. The initial state of the membrane ("UHV" curve at the pressures of both H_2S and O_2 less than 10^{-8} Pa) obtained as a result of 25 h bombardment by 2 keV ion beam 3×10^{14} cm^{-2} s^{-1}.

the membrane permeability to ions, as seen in fig. 5, by a factor of about 10^3, with χ reaching up to about 0.2. At lower temperatures the effect of O_2 is not so large (fig. 6), but the presence of H_2S results in a dramatic increase in χ and c, and in a decrease in α_0 and k_{r0} as well *, when under the action of ions [64], as shown in figs. 6–8.

* The quantities α_0 and k_{r0} have been deduced as follows: First, the χ values were measured for thermal atoms and for ions (dots in figs. 6 and 7) as well as for thermal molecules. The latter was found to be independent of the driving pressure, p, before initiating the ion beam, while for the transient time, τ, was $\tau > L^2/6D$. Consequently: (i) $\chi \approx \alpha/2$ (eq. (10c)) for H_2 molecules, as long as $\alpha_0 \approx \alpha_L$ before the ion bombardment, and (ii) the concentration, c, does not depend on coordinate. Taking this into account, one will get $\alpha_L = 2\chi = 8\times10^{-3}\exp(-39\ \text{kJ mol}^{-1}/RT)$, and $k_{rL} = \alpha_L Z/S^2 = 1.5\times10^{-18}\exp(-71.5\ \text{kJ mol}^{-1}/RT)$, from the experimental data for χ, whence c may be computed according to the equation $k_{rL}c = \chi j_i$; the data for c are presented in figs. 6 and 7. Now one can prove, inserting α_L in eqs. (4), and using the data from table 1, that eq. (4a) is valid over the temperature range investigated and at the given values of L and j_i for the "hot" gas. Considering $\alpha_0 \gg \alpha_L$ after the ion bombardment, one gets $\chi = \xi\alpha_L/\alpha_0$. Hence α_0 and $k_{r0} = \alpha_0 Z/S^2$ are derived and plotted in fig. 8 (dependence of ξ on ion energy E was accounted for by assuming [65] $\xi = 0.2$, 0.4, 0.7, 1.0 at $E = 10$, 10^2, 10^3, 5×10^3 eV, respectively).

One is tempted to single out the left branches of the curves in fig. 6a, as the permeability drops by orders of magnitude when the ion energy, E, rises; this happens despite of the fact that the depth of implantation increases with E, and the reflection coefficient for ions decreases. A likely explanation presents itself, prompted by the deep minimum in the χ versus E plots corresponding to the E range where a maximum in the sputtering yield is usually observed [66]. Apparently, the large-scale and non-monotonic varying of χ (and hence of k_{r0} and c) with E must be controlled by the degree of compactness of the sulphur overlayer on the metal surface, which depends on the effectiveness of ion sputtering. The more sulphur deposited per unit time, the denser the overlayer at a given sputtering rate. Hence, the increase in χ (and in c) with H_2S pressure and temperature (figs. 6 and 7). For thermal atoms (dots at $E = 2.5$ eV in fig. 6: ascribed to thermal atoms is the energy released when the atom is bound by the metal, viz., 2–3 eV), χ depends neither on H_2S pressure nor on metal temperature even at the lowest H_2S pressure investigated (figs. 6 and 7). This is understood on assuming that the sulphur adlayer on the stainless steel surface is stable when exposed to atomic hydrogen, just as it occurs in the case of S on Pd (section 4.2). It is also understandable why the energy dependence of χ and c in fig. 6 is more pronounced at lower H_2S pressures and at lower temperatures. When the rate of deposition of

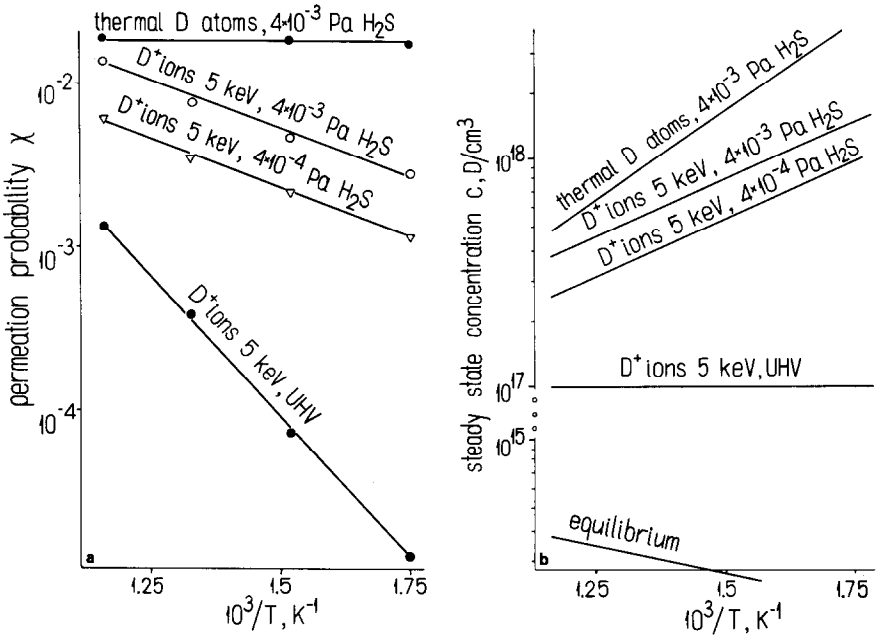


Fig. 7. Effect of H_2S present on upstream side of an austenitic stainless steel membrane upon temperature dependence of permeability (experimental) and upon steady state hydrogen concentration in metal (derived from the permeability data). The incident flux j_i is 3×10^{14} cm^{-2} s^{-1}.

sulphur is low, the amount of sulphur on the metal surface may vary from a compact overlayer to almost a free surface, depending on the rate of sputtering.

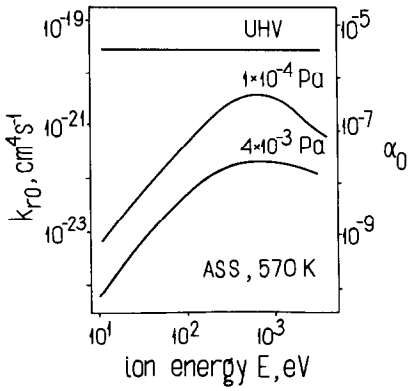


Fig. 8. Presented here as functions of acting ions energy E are: k_{r0}, the rate constant for associative thermal re-emission of hydrogen by austenitic stainless steel, and α_0, the probability of H_2 molecule *thermal* dissolution in austenitic stainless steel, *when under* D^+ *ion bombardment* (incident flux 3×10^{14} D^+/cm^2 s), in ultrahigh vacuum (after 23 h bombardment by 2 keV D^+ ions – cf. fig. 5), and in the presence of H_2S (1×10^{-4} and 4×10^{-3} Pa). The plots are deduced from the permeability data (fig. 6).

If one had been considering any single non-monotonic plot of χ versus E or c versus E in fig. 6, one might conclude that their peculiar pattern of behaviour should be put down to the stimulated evolution (self-sputtering) of the implanted hydrogen. That such an inference would have been wrong is clearly evidenced by the fact that, at each specific ion energy, the ion driven permeability depends on H_2S pressure on the upstream side and on metal temperature. The ion sputtering affects the gas evolution indirectly, by cleaning the surface and thus enhancing the thermal evolution rate constant k_{r0} (see fig. 8).

The effect of oxygen on the energy dependence differs from that of hydrogen sulphide, as seen in fig. 6. A reason for this may be that sulphur does not form more than a monatomic adlayer, while thick oxide films are possible following treatment in oxygen at 570 K. Thermal hydrogen atoms and low energy ions easily penetrate the sulphur monolayer, as in the case of S on Pd. But their energy is insufficient to overcome thick oxyde coats at first impact. The low energy particles are stopped in the oxide film and have to diffuse inside it. The probability of the dissolution in metal increases with the depth of implantation until the ions enter directly into the metal lattice.

It has been demonstrated recently in the direct experiments [60,67] that the drop with time of the ion driven permeability of stainless steel should be due to cleaning the surface of specific impurities, such as oxygen and sulphur, under the action of ion sputtering.

Summing up, experiments on ASS membranes reveal the large-scale effects of rather delicate surface chemistry on hydrogen permeation, uptake and re-emission under non-equilibrium conditions. Addressing the engineering issues of fusion device technology, the chemical composition of the first wall surface and of in-vessel components will be defined by the integral set of physico-chemical conditions. These include the specific materials of component fabrication, the spectrum of residual gases, the purity of the fuel mixture, the fluxes of energy and particles falling upon the surface, the metal temperature, etc.

Austenitic stainless steel is hardly a suitable material to attain superpermeability because of the very high barrier $\mathscr{E}_{a3}$ (see eq. (9), fig. 1 and table 1). That superpermeability can be reached, or at least approached, with ASS membranes demonstrates the immense opportunities that manipulating the surface chemistry may render to control permeability, uptake and re-emission.

5.2. Niobium

The metals of Group Va, vanadium, niobium and tantalum, are much more suitable for designing systems to pump out, purify and compress the fuel in fusion devices. The barrier $\mathscr{E}_{a3}$ is well below zero for these metals (see fig. 1 and table 1). It follows, as seen from eq. (9), that superpermeability is possible at likely membrane thicknesses, L, and incident fluxes, j_i, even when both the membrane surfaces are clean. Still more promising appear the membranes of these metals with their surfaces passivated, particularly, to the favourable side (i.e. the upstream surface passivated stronger than the downstream one).

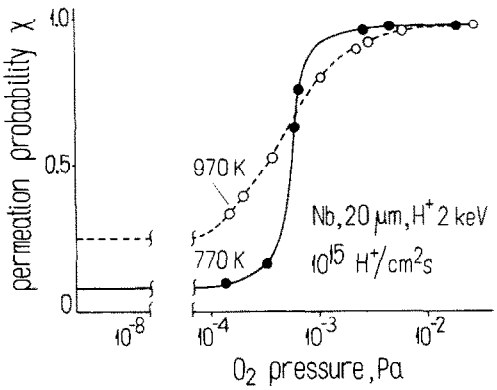


Fig. 9. Effect of oxygen present on upstream side of a niobium membrane upon the membrane permeability to H^+ ions. Initially the membrane had had a symmetric passivation ($\alpha_0 \approx \alpha_L$) due to incomplete cleaning. H^+ bombardment resulted in $\alpha_0 > \alpha_L$, and consequently χ depends on temperature, and $\chi < 0.5$ (left section of the plot). The admittance of oxygen turns the asymmetry "upside down": becomes $\alpha_0 \ll \alpha_L$ and $\chi = \text{const}(T) \approx 1$.

We began our experiments on niobium membranes with both surfaces passivated due to incomplete cleaning. Superpermeability was observed [45,47] over the

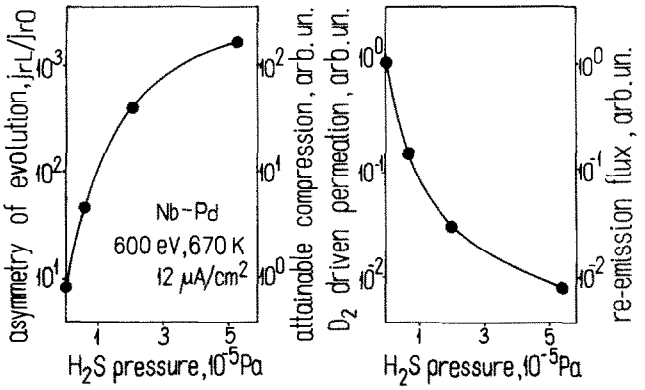


Fig. 10. Effect of H_2S present on upstream side upon the asymmetry of evolution of implanted hydrogen by, and molecular hydrogen driven permeation through, a 100 μm-thick niobium membrane with palladium overlayer deposited on the downstream surface (experimental). Also drawn are the scales for the attainable rate of compression and for re-emission flux, when under deuterium ion beam bombardment (computation from these experimental data).

whole range of metal temperatures (570–1370 K) and incident fluxes (up to $\sim 10^{16}$ cm^{-2} s^{-1}) investigated for thermal atoms and for ions (with energy ranging from 30 eV to 7 keV). This holds as long as the upstream surface remains passivated by the interaction with gases present on the upstream side of the membrane: ambient residual gases of the vacuum system or gases which appear when the hydrogen ion beam is initiated (e.g. water vapour [45]). Once this gas pressure becomes insufficient to maintain a continuous passive layer, a membrane, originally symmetric, becomes non-symmetric ($\alpha_0 \gg \alpha_L$) because of contaminants sputtered from the upstream surface. This asymmetry may be compensated, and even turned "upside down" by admitting oxygen (fig. 9) or hydrogen sulphide onto the upstream side [68]. Thus $\chi \approx \xi \approx 1$ can be attained, and also maintained during the ion bombardment. That χ is indeed close to unity is evidenced by direct measurements and by the fact that χ does not rise further when O_2 or H_2S pressure is increased further (fig. 9).

Coating the downstream surface by a palladium overlayer after cleaning results in asymmetry beneficial to superpermeability ($\alpha_L \gg \alpha_0$) even when the upstream niobium surface is clean, since intrinsically $\alpha_{Pd} > \alpha_{Nb}$, as demonstrated by Pick et al. [69]. The passivation of upstream surface is still advantageous because it entails an additional orders-of-magnitude decrease in re-emission when the membrane is exposed to hydrogen ions, as shown in fig. 10. Simultaneously, the permeability to molecular hydrogen is radically suppressed. The attainable degree of compression is to increase by orders of magnitude, alongside of a tolerable amount of the quantity $L\sqrt{j_i}$ for which superpermeability will hold according to eq. (9), as a result of the upstream surface passivation.

6. Models for "hot" hydrogen interaction with walls and components of fusion devices – a physico-chemical approach

The data presented above suggest that the dynamics of surface chemical composition may govern permeation, uptake and reemission of hydrogen when a metal is acted upon by "hot" gas. The dynamic sputtering of surface contaminants and their dissolution in the bulk material is counterbalanced by chemical reactions with gases present in the vacuum chamber and by surface segregation of contaminants from the bulk. Thus the characteristics of upstream surface, such as α_0, σ_0 and $(\mathscr{E}_{a1})_0$ in eqs. (4)–(11), will depend upon the whole set of physical and physico-chemical conditions existing in each particular device or assembly [7,57]. Namely, one should consider the energy and flux of incident particles, temperature, the composition of residual gases (cf. fig. 8), the material of in-vessel components (because of plasma driven transport of matter), and the initial concentration of contaminants on the surface and in the bulk of the material. Of crucial importance might be the danger of vacuum leaks which, if beginning suddenly, are likely to bring about dramatic changes in hydrogen uptake and permeation.

Uptake, recycling and permeation becomes rather a complicated problem, and, in particular, if the barrier $\mathscr{E}_{a1}$ depends on incident particles energy E, incident flux j_1 and metal temperature. Such a dependence is not necessarily the case; for instance, in fig. 7 the plots presenting the temperature dependence of permeability at two different pressures of H_2S are parallel and differ only in the pre-exponential factor σ. A physical picture may be that the less the rate of sulphur deposition, the larger the number of facilitated re-emission centres (holes in sulphur overlayer) produced by ion bombardment, with the same barrier $(\mathscr{E}_{a1})_0$ being characteristic of all these centres.

A number of physical models, transport parameter formalisms and computer codes have been developed to evaluate hydrogen uptake, recycling and permeation in fusion devices. However, even for the most sophisticated and multi-variant treatments, the question remains what values should be taken for the kinetic parameters. More or less accurate information is available for the bulk material properties, such as equilibrium solubility, diffusion, heat transport, etc., but that is not so for the kinetics of surface processes. A vast volume of data concerning the interaction of hydrogen with clean metal surfaces is of little help here. Experiments with occasional surface passivation do serve to demonstrate the utmost importance of the surface chemistry, though they may have little relation to *real surfaces* in fusion devices. One of the aims of the present paper is to emphasize the salient necessity that comprehensive assessments of the physico-chemical conditions in fusion devices be made and taken into account. Unless this is done, even qualitative estimates of hydrogen uptake and permeation will be unreliable.

7. Suggestions to use superpermeability in fusion devices

We would like to interest those involved in the design of fusion reactors in endeavoring to make use of superpermeability, since it is, in our opinion, not just a curious "exotic" effect, but a real mechanism that can

be effectively employed to pump, to separate hydrogen from helium and to recuperate hydrogen isotopes.

One straightforward idea is a superpermeable membrane to replace the burial targets for neutral beam injectors. The passivating films providing for superpermeability will be sputtered by fast particles from the inlet target surface, but maintaining these films dynamically, or renewing them periodically, will not raise major difficulties. Non-metallic films may be re-deposited by interaction with chemically active gases, as shown in this paper. One can use, among others, carbon-containing gases, such as acetylene. The probability of decomposition of a gas molecule at single impact on a metal surface is often close to unity (e.g. for C_2H_2 on Nb [70,71]). Thus the admitted gas flow may be used up with a great efficiency, minimizing essentially the contamination of plasma by the influx of admitted gas from the target chamber into the main vacuum chamber of a fusion device, particulary, if the gas is supplied to the target surface in the form of a directed stream.

Another option is to evaporate carbon, or the metals that possess a natural $\mathscr{E}_{a1}$ barrier characteristic of their clean surfaces (e.g. copper [45,54], gold [72], perhaps aluminium, etc.), onto the inlet target surface. The deposition of surface passivating films would seem advantageous even regarding the more traditional concept of burial target. Such films might improve markedly the target pumping capacity.

Superpermeable membranes may be used immediately as pumping targets for the injectors in mirror fusion devices of these days (e.g. refs. [73] and [74]). As for the injectors being designed for large tokamak machines, the problem arises of heat load on the target. This difficulty might be avoided though, if the energetic particles were retarded to recycle the energy. The heat may be also conducted away by a liquid metal plenum contacting the membrane target on the outlet side. If alkali or alkaline earth metals are used which are good hydrogen occluders, the implanted hydrogen will pass through the membrane into the liquid metal [38] and will be driven away by circulating melt. The dissolved hydrogen isotopes may then be extracted from the melt by heating, and returned to the fusion cycle. Considering the heat load problem, it is again fortunate that such refractory metals as vanadium, niobium and tantalum are most suitable to make superpermeable membranes of them.

One could use superpermeable membranes for the evacuation of hydrogen isotopes from fusion devices without exposing the membranes to the power load from primary corpuscular fluxes. For example, hydrogen might be separated from helium in the tokamak divertor pumping ducts by superpermeable membranes placed along the duct walls lengthwise. The temperature of the divertor collection plate being ~ 2000 K, the plate will produce the flow of hydrogen thermal atoms onto the duct walls alongside of reflected fast hydrogen particles.

Yet another place were superpermeable membranes may be used to pump away non-equilibrium hydrogen is plasma sources of hydrogen particle beams, especially, on the earlier stages of acceleration when the energy of diffused particles is not very high.

We also deem it feasible to employ a superpermeable membrane as a pumping liner in mirror fusion devices. The vacuum limitations are much more severe there than in tokamaks for plasma performance. Designed passivation of the surface of pumping liner will secure the low rate of gas recycle from the wall, so as not to degrade the plasma during burn pulse, and such passivation appears to be necessary for the both options of sorptional [75] and membrane pumping by the liner.

Finally, it is possible to design separate pump units containing their own source of "hot" hydrogen and employing superpermeability. The model devices of that kind have been built [32,48], using hydrogen atomization on an incandescent tungsten filament.

Appendix. Table of notation

L = membrane thickness,
ξ = the probability of retention of a "hot" gas particle by metal (per single impact on the surface),
α = $\alpha^*\sigma \exp(-2\mathscr{E}_{a1}/RT)$ = the probability of dissociative binding of a thermal hydrogen molecule by metal (per single impact on the surface),
σ = the portion of a surface occupied by the centres active in the processes of hydrogen thermal dissolution/evolution,
χ = permeation probability (per single impact on a membrane surface),
j_i = incident flux of hydrogen particles,
j_r = the flux of hydrogen being evolved by a metal surface,
$j_{rL} = j_i\chi$ = permeation flux,
p = molecular hydrogen pressure,
c = concentration of hydrogen dissolved in metal,
$c_e = S\sqrt{p}$ = equilibrium concentration of dissolved hydrogen,
$S = S^* \exp(-\Delta H/RT)$ = Sieverts' constant,
$k_r = Z\alpha/S^2$ = rate constant for hydrogen thermal evolution,

$Z = (2\pi mkT)^{-1/2}$ = kinetic theory coefficient ($Z \approx 1.2 \times 10^{19} \mathrm{cm}^{-2}\ \mathrm{s}^{-1}\ \mathrm{Pa}^{-1}$ for H_2 at room temperature),

$D = D^*\ \exp(-\mathscr{E}_D/RT)$ = diffusion coefficient for dissolved hydrogen.

The subscripts 0 and L refer to a membrane inlet (upstream) and outlet (downstream) sides respectively, the superscript * denotes pre-exponential factors of the corresponding quantities. The energetic quantities $\mathscr{E}_{a1}$, $\mathscr{E}_{a3}$, $\mathscr{E}_D$ and ΔH are explicated in fig. 1.

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References

[1] A.I. Livshits, Sov. Tech. Phys. Lett. 3 (1977) 236.
[2] V.R. Kapitanskiy, A.I. Livshits and I.M. Metter, patent registered with the Soviet Scientific Authority (I.Cl. H01s 1/06, No 437167, reported 26 April 1971, published 25 July 1974).
[3] V.R. Kapitanskiy, A.I. Livshits and I.M. Metter, patent registered with the Soviet Scientific Authority (I.Cl. C01b 1/27, No 446466, reported 26 April 1971, published 15 October 1974).
[4] A.I. Livshits, Sov. Phys. Tech. Phys. 21 (1976) 187.
[5] A.I. Livshits, M.E. Notkin, Yu.M. Pustovoit and A.A. Samartsev, Hydrogen Absorption and Permeation in Metals under the Non-Equilibrium Conditions, Preprint 3516/8 (Kurchatov Atomic Energy Institute, Moscow, 1981) in Russian; translated into English in: INTOR, Phase Two A, Conclusion from the 4th Workshop Meeting EUR FU, Brussels, 1982.
[6] F. Waelbroeck, Influence of Bulk and Surface Phenomena on the Hydrogen Permeation through Metals, Report JÜL-1966, Kernforschungsanlage Jülich GmbH, Jülich (1984).
[7] A.L. Livshits, Topics in Atomic Science and Technology, Ser.: Thermonuclear Fusion (in Russian) 4 (1987) 35; 1 (1988) 69.
[8] I.V. Altovskiy, M.V. Zhuk and V.V. Pistunovich, At. Energiya 63 (1987) 20, in Russian; translated into English in: Sov. At. Energy.
[9] M.W. Roberts and C.S. McKee, Chemistry of the Metal–Gas Interface (Clarendon Press, Oxford, 1978).
[10] V.R. Kapitanskiy, A.I. Livshits, I.M. Metter and M.E. Notkin, Sov. Phys. Tech. Phys. 21 (1976) 602.
[11] A.I. Livshits, I.M. Metter and A.A. Samartsev, Sov. Phys. Tech. Phys. 21 (1976) 848.
[12] A.I. Livshits, M.E. Notkin and S.V. Yakovlev, Sov. Tech. Phys. Lett. 4 (1978) 192.
[13] A.I. Livshits, M.E. Notkin, Yu.M. Pustovoit and A.A. Samartsev, Interaction of Thermal Atomic and Molecular Hydrogen with Metallic Membranes, Preprint 3517/8 (Kurchatov Atomic Energy Institute, Moscow, 1981) in Russian; translated into English in: INTOR, Phase Two A, Conclusion from the 4th Workshop Meeting EUR FU, Brussels, 1982.
[14] A.Yu. Doroshin, A.I. Livshits and A.A. Samartsev, Poverkhn. Fiz. Khim. Mekh. No 8 (1985) 31, in Russian; translated into English in: Phys. Chem. Mech. Surf. 4 (1987) 2321.
[15] A.Yu. Doroshin, A.I. Livshits and A.A. Samartsev, Khim. Fiz. 7 (1988) 841, in Russian; to be translated into English in: Sov. J. Chem. Phys.
[16] A.I. Livshits, Vacuum 29 (1979) 103.
[17] Gase und Kohlenstoff in Metallen, Eds. E. Fromm and E. Gebhardt (Springer, Berlin, 1976).
[18] INTOR, Phase Two A, Part One (IAEA, Vienna, 1983) pp. 327–347.
[19] A.Yu. Doroshin, A.I. Livshits and A.A. Samartsev, Poverkhn. Fiz. Khim. Mekh. (The Surface: Physics, Chemistry and Mechanics) No. 7 (1988) 841, in Russian.
[20] A.I. Livshits and A.A. Samartsev, Sov. Phys. Tech. Phys. 49 (1979) 1365.
[21] S.M. Ko and L.D. Schmidt, Surf. Sci. 42 (1979) 508.
[22] M. Pick, Phys. Rev. B 24 (1981) 4287.
[23] F. Waelbroeck, I. Ali-Khan, K.J. Dietz and P. Wienhold, J. Nucl. Mater. 85 & 86 (1979) 345.
[24] V.B. Kazanskiy, R.E. Mardaleishvili, V.P. Strunin and V.V. Voyevodskiy, Zh. Fiz. Khim. 30 (1956) 821, in Russian.
[25] A.I. Livshits, Sov. Phys. Tech. Phys. 20 (1975) 1207.
[26] M.A. Pick and K. Sonnenberg, J. Nucl. Mater. 131 (1985) 208.
[27] M.A. Pick, J. Nucl. Mater. 145–147 (1987) 297.
[28] A.A. Pisarev, V.G. Telkovskiy and V.N. Tsyplakov, Poverkhn. Fiz. Khim. Mekh. No 6 (1985) 49, in Russian; translated into English in: Phys. Chem. Mech. Surf. 4 (1986) 1701.
[29] I. Ali-Khan, F. Waelbroeck, K.J. Dietz and P. Wienhold, J. Nucl. Mater. 76 & 77 (1978) 337.
[30] M.A. Krykin and S.F. Timashov, Sov. Phys. – Solid State 27 (1985) 1529.

[31] A.A. Ovchinnikov, S.F. Timashov and A.A. Bely, Kinetics of the Diffusion-Controlled Chemical Processes (Khimiya, Moscow, 1986) in Russian.
[32] A.I. Livshits, M.E. Notkin, Yu.M. Pustovoit and A.A. Samartsev, Vacuum 29 (1979) 113.
[33] A.I. Livshits and Yu.M. Pustovoit, in: INTOR, Phase Two A, USSR Contribution to the 5th Workshop Meeting (IAEA, Vienna, 1983) p. 17.
[34] N.I. Ionov, Sov. Phys. – Solid State 27 (1985) 1527.
[35] W.A. Swansiger and D.R. Folk, Rev. Sci. Instr. 58 (1987) 1266.
[36] A.A. Samartsev, Kinet. Catal. 26 (1985) 703.
[37] W.A. Swansiger and R. Bastasz, J. Nucl. Mater. 85 & 86 (1979) 335.
[38] F. Behr, R. Schulten and W. Weirich, USA Patent (I.Cl. B01D 53/22, No 4496373, published 29 January 1985).
[39] G. Comsa and R. David, Surf. Sci. 117 (1982) 77.
[40] G. Comsa and R. David, Surf. Sci. Rep. 5 (1985) 145.
[41] A. Günterschulze, H. Betz and H. Kleinwächer, Zs. f. Phys. 111 (1938) 657.
[42] H. Betz, Zs. f. Phys. 117 (1940) 100.
[43] J.F. Lynch and T.B. Flanagan, Surf. Sci. 62 (1977) 519.
[44] E. Wicke and H. Brodowsky, in: Topics in Applied Physics, Vol. 29, Hydrogen in Metals II, Application-Oriented Properties, Ed. G. Alefeld and J. Völkl (Springer, Berlin, 1978).
[45] A.I. Livshits and M.E. Notkin, Sov. Tech. Phys. Lett. 7 (1981) 605.
[46] A.I. Livshits, M.E. Notkin, Yu.M. Pustovoit and S.V. Yakovlev, Topics in Atomic Science and Technology, Ser.: Thermonuclear Fusion (in Russian) 2 (10) (1982) 73.
[47] A.I. Livshits, M.E. Notkin, S.V. Yakovlev and Yu.M. Pustovoit, ibid. ref. [46], p. 77.
[48] H. Hackfort, K. Bösche, F. Waelbroeck, J. Winter and P. Wienhold, J. Nucl. Mater. 144 (1987) 10.
[49] G. Comsa, R. David and B.-J. Schumacher, Surf. Sci. 85 (1978) 45.
[50] T. Engel and H. Kuipers, Surf. Sci. 90 (1979) 162.
[51] H.K. Perkins and T. Noda, J. Nucl. Mater. 71 (1978) 349.
[52] R.A. Kerst and W.A. Swansiger, J. Nucl. Mater. 122 & 123 (1984) 1499.
[53] A.Ya. Tontegode, D.Sc. Thesis, Ioffe Physical Engineering Institute, USSR Academy of Science, Leningrad (1974).
[54] N.R. Gall, S.M. Mikhailov, E.V. Rutkov and A. Ya. Tontegode, Surf. Sci. 191 (1987) 185.
[55] E.Ya. Zandberg, A. Ya. Tontegode and F.K. Yusifov, Sov. Phys. Tech. Phys. 17 (1972) 134.
[56] Adsorption on Metal Surfaces: An Integrated Approach, Ed. J. Bénardt (Elsevier, Amsterdam, 1983).
[57] A.I. Livshits, Yu.M. Pustovoit and V.S. Svishchov, Topics in Atomic Science and Technology, Ser.: Thermonuclear Fusion (in Russian) 2 (8) (1981) 45.
[58] A.Yu. Doroshin, Candidate Science Thesis, Surface and Vacuum Research Centre, USSR State Commitee of Standards, Moscow (1988).
[59] A.I. Livshits, M.E. Notkin and Yu.M. Pustovoit, Topics in Atomic Science and Technology, Ser.: Thermonuclear Fusion (in Russian) 2 (15) (1984) 64.
[60] R.A. Causey, R.A. Kerst and B.E. Mills, J. Nucl. Mater. 122 & 123 (1984) 1547.
[61] A.A. Kurdyumov, Yu.I. Belyakov and Yu.I. Zvezdin, Sov. Phys. Tech. Phys. 18 (1973) 106.
[62] J. Winter, F. Waelbroeck, P. Wienhold and T. Schelske, J. Nucl. Mater. 111 & 112 (1982) 243.
[63] T. Tanabe, N. Saito, Y. Etoh and S. Imoto, J. Nucl. Mater. 103 & 104 (1981) 483.
[64] A.I. Livshits, M.E. Notkin and A.A. Boikov, Topics in Atomic Science and Technology, Ser.: Thermonuclear Fusion (in Russian) 3 (1986) 56.
[65] V.A. Kurnayev, E.S. Mashkova and V.A. Molchanov, Reflection of Light Ions by Solid Surfaces (Energoatomizdat, Moscow, 1985) in Russian.
[66] Topics in Applied Physics, Vol. 47, Sputtering by Particle Bombardment. I. Physical Sputtering of Single-Element Solids, Ed. R. Behrisch (Springer, Berlin, 1981).
[67] M. Yamawaki, K. Yamaguchi, T. Kigoshi and T. Namba, J. Nucl. Mater. 145–147 (1987) 309.
[68] A.I. Livshits, D.Sc. Thesis, Kurchatov Atomic Energy Institute, Moscow (1982).
[69] M.A. Pick, M.G. Greene and M. Strongin, J. Less-Comm. Met. 73 (1980) 89.
[70] G. Hörz and K. Lindenmaier, J. Less-Comm. Met. 35 (1974) 85.
[71] G. Hörz, Metall. 27 (1973) 680.
[72] A.A. Kurdyumov, to be published in Fiz.-Khim. Mekh. Mater. (in Russian), translated into English as Sov. Mater. Sci.
[73] G.I. Dimov and G.V. Roslyakov, A Trap with Ambipolar Plugs, Preprint 80–152 (Nuclear Physics Institute, USSR Academy of Science Siberian Division, Novosibirsk, 1980).
[74] E.D. Bender, Topics in Atomic Science and Technology, Ser.: Thermonuclear Fusion (in Russian) 4 (1987) 41.
[75] G.I. Dimov, Topics in Atomic Science and Technology, Ser.: Thermonuclear Fusion (in Russian) 3 (1988) 13.